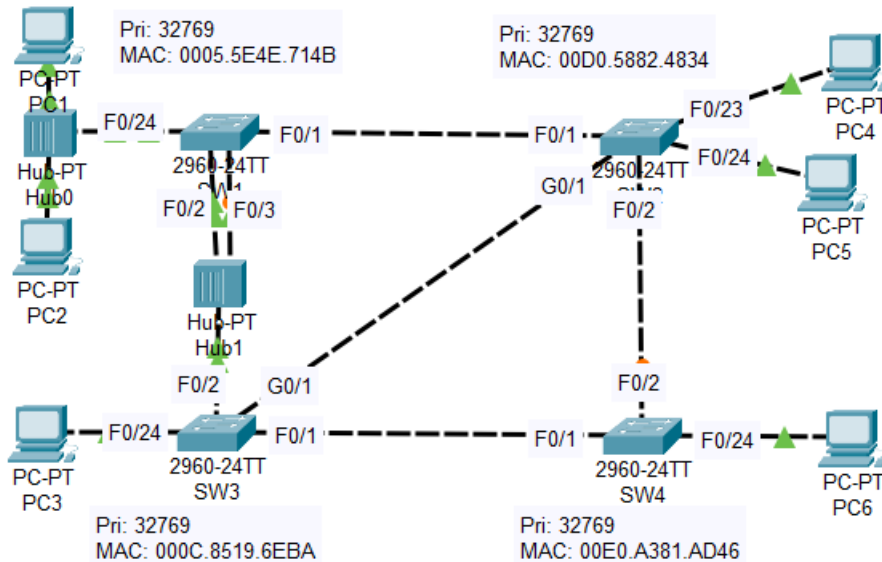




Topology

1. Which switch is the root bridge? Use the CLI to examine the port role/state of each interface on the root. What appears different than what you have learned about the root bridge? What is the cause of this?

They all have the same bridge priority which means they belong to the same VLAN. SW1 has the lowest MAC address hence it is the root bridge.

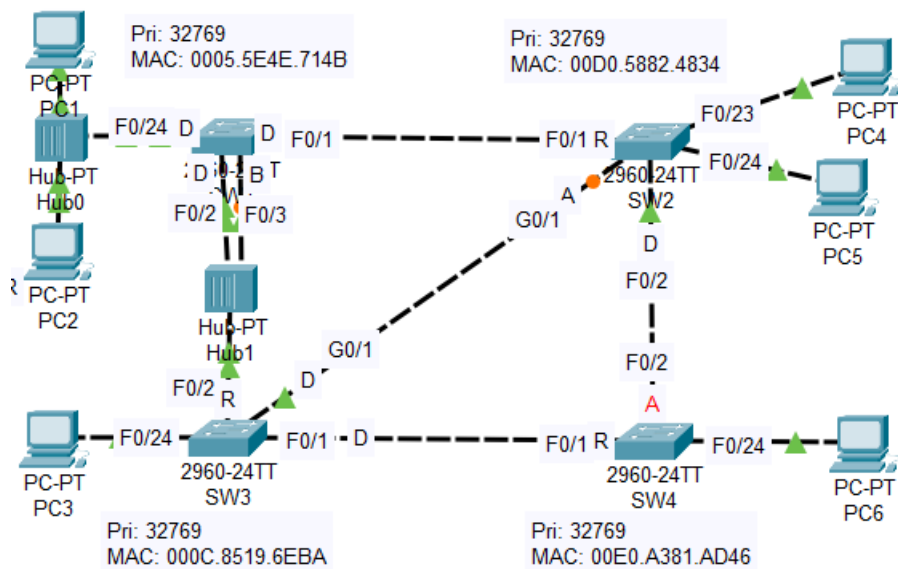
```
SW1>en
SW1#show spanning-tree
VLAN0001
  Spanning tree enabled protocol rstp
  Root ID    Priority    32769
            Address      0005.5E4E.714B
            This bridge is the root
            Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
            Address      0005.5E4E.714B
            Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
            Aging Time 20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/2	Desg	FWD	19	128.2	Shr
Fa0/3	Back	BLK	19	128.3	Shr
Fa0/24	Desg	FWD	19	128.24	Shr
Fa0/1	Desg	FWD	19	128.1	P2p

F0/2, F0/1 and F0/24 are designated ports, while F0/3 is a backup port in a discarding state, written as blocking in the CLI output.

2. Without using the CLI, determine the port role/state of each remaining switch interface. Use the CLI to confirm.



```

SW2>en
SW2#show spanning-tree
VLAN0001
  Spanning tree enabled protocol rstp
    Root ID    Priority    32769
              Address      0005.5E4E.714B
              Cost        19
              Port        1(FastEthernet0/1)
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

    Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
              Address      00D0.5882.4834
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  20
  
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/2	Desg	FWD	19	128.2	P2p
Fa0/1	Root	FWD	19	128.1	P2p
Fa0/23	Desg	FWD	19	128.23	P2p
Fa0/24	Desg	FWD	19	128.24	P2p
Gi0/1	Altn	BLK	4	128.25	P2p

```

SW3>en
SW3#show spanning-tree
VLAN0001
  Spanning tree enabled protocol rstp
    Root ID    Priority    32769
              Address      0005.5E4E.714B
              Cost         19
              Port         2(FastEthernet0/2)
              Hello Time   2 sec   Max Age 20 sec   Forward Delay 15 sec

    Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
              Address      000C.8519.6EBA
              Hello Time   2 sec   Max Age 20 sec   Forward Delay 15 sec
              Aging Time   20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/1          Desg FWD 19        128.1    P2p
Fa0/2          Root FWD 19        128.2    Shr
Fa0/24         Desg FWD 19        128.24   P2p
Gi0/1          Desg FWD 4        128.25   P2p

SW4>en
SW4#show spanning-tree
VLAN0001
  Spanning tree enabled protocol rstp
    Root ID    Priority    32769
              Address      0005.5E4E.714B
              Cost         38
              Port         1(FastEthernet0/1)
              Hello Time   2 sec   Max Age 20 sec   Forward Delay 15 sec

    Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
              Address      00E0.A381.AD46
              Hello Time   2 sec   Max Age 20 sec   Forward Delay 15 sec
              Aging Time   20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/1          Root FWD 19        128.1    P2p
Fa0/2          Altn BLK 19        128.2    P2p
Fa0/24         Desg FWD 19        128.24   P2p

```

3. Manually configure the appropriate RSTP link type on each interface. What do you think is the correct link type for SW1's F0/24?

```

SW2(config)#int range f0/23-24
SW2(config-if-range)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/23 but will only
have effect when the interface is in a non-trunking mode.
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/24 but will only
have effect when the interface is in a non-trunking mode.

```

```
SW1>en
SW1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW1(config)#do sh span
VLAN0001
  Spanning tree enabled protocol rstp
  Root ID      Priority    32769
              Address     0005.5E4E.714B
              This bridge is the root
              Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID    Priority    32769 (priority 32768 sys-id-ext 1)
              Address     0005.5E4E.714B
              Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time 20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/2	Desg	FWD	19	128.2	Shr
Fa0/3	Back	BLK	19	128.3	Shr
Fa0/24	Desg	FWD	19	128.24	Shr
Fa0/1	Desg	FWD	19	128.1	P2p

```
SW1(config)#int range f0/24
SW1(config-if-range)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION
```

%Portfast has been configured on FastEthernet0/24 but will only have effect when the interface is in a non-trunking mode.

F0/24 is a shareport and edge port.

```
SW3#show spanning-tree
VLAN0001
  Spanning tree enabled protocol rstp
  Root ID      Priority    32769
              Address     0005.5E4E.714B
              Cost        19
              Port        2(FastEthernet0/2)
              Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID    Priority    32769 (priority 32768 sys-id-ext 1)
              Address     000C.8519.6EBA
              Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time 20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Desg	FWD	19	128.1	P2p
Fa0/2	Root	FWD	19	128.2	Shr
Fa0/24	Desg	FWD	19	128.24	P2p
Gi0/1	Desg	FWD	4	128.25	P2p

```
SW3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW3(config)#int f0/24
SW3(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION
```

%Portfast has been configured on FastEthernet0/24 but will only have effect when the interface is in a non-trunking mode.

```
SW4(config)#int range f0/1-2
SW4(config-if-range)#spanning-tree link-type point-to-point
SW4(config-if-range)#int f0/24
SW4(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION
```

```
%Portfast has been configured on FastEthernet0/24 but will only
have effect when the interface is in a non-trunking mode.
```

```
SW4(config-if)#do sh span
```

```
VLAN0001
```

```
Spanning tree enabled protocol rstp
```

```
Root ID      Priority    32769
             Address     0005.5E4E.714B
             Cost        38
             Port        1(FastEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
```

```
Bridge ID    Priority    32769 (priority 32768 sys-id-ext 1)
             Address     00E0.A381.AD46
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Root	FWD	19	128.1	P2p
Fa0/2	Altn	BLK	19	128.2	P2p
Fa0/24	Desg	FWD	19	128.24	P2p

F0/24 is still a P2p type even though PortFast was enabled on it. Edge ports connected to an end host like this will still be point to point links if they are using full duplex. There is no hub here, therefore the link is full-duplex. However, PortFast is still enabled.